

YUVARAJ S

B.Tech Artificial Intelligence and Data Science



yuvarajselvam2006@gmail.com



+91 8248546976



<https://www.linkedin.com/in/yuvaraj-s-3b28442bb/>



<https://github.com/Yuvarajsel>

CAREER OBJECTIVE

Motivated and detail-oriented B.Tech Artificial Intelligence and Data Science student with strong foundations in programming, problem-solving, data structures, and software development. Seeking an opportunity to apply technical skills, contribute to innovative projects, and gain industry experience in a dynamic organization.

EDUCATION

HIGHERSECONDARY EDUCATION: 2022-2023 | Score:69%

Government Higher Secondary School, Vathalmalai, Dharmapuri

BACHELOR OF TECHNOLOGY: 2023-2027 | CGPA: 6.52

Bannari Amman Institute of Technology, Sathyamangalam, Erode

TECHNICAL SKILLS:

Programming Languages: Python, Java, C

Web Technologies: HTML5, CSS3, JavaScript, Flask, REST API Development

Databases: MongoDB, MySQL

Core Computer Science: Data Structures and Algorithms, Object-Oriented Programming, DBMS

AI / ML: Machine Learning, RAG Pipelines, Prompt Engineering, NLP, Computer Vision

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

PROJECT EXPERIENCE:

Voice-Based AI Study Assistant

- Developed a voice-enabled AI study assistant capable of analyzing spoken input and extracting study topics automatically.
- Implemented interactive quiz and assessment features to improve learning engagement.
- Enabled real-time audio communication using speech processing techniques.

Visual Search Engine for Jewelry using Multimodal RAG

- Designed a multimodal retrieval system allowing search through images, sketches, and text inputs.
- Implemented vector similarity search for accurate jewelry item retrieval.
- Generated meaningful product descriptions using Large Language Models (LLMs).

Plant Growth Prediction System

- Developed a machine learning model to predict plant growth using environmental and soil parameters.
- Applied data preprocessing and feature engineering techniques to improve prediction accuracy.
- Evaluated model performance using regression algorithms.

CropCopilot – AI-Powered Agricultural Intelligence System

- Developed an AI-powered agriculture intelligence system combining RAG and Text-to-SQL for smart crop analysis.
- Built the system using NVIDIA NIM APIs and CrewAI agents to retrieve agronomic knowledge and query structured farm data.
- Designed a FastAPI backend with a modern web interface to deliver actionable agricultural insights in real time.

LANGUAGES KNOWN

- English
- Tamil